Regression Evaluation Metrics

Model Evaluation using MSE, RMSE, MAE

Mean Square Error(MSE):

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2$$

MSE = mean squared error

n = number of data points

Y_i = observed values

$\hat{Y}_i$ = predicted values

- It is a technique to measure the average square error of a regression algorithm.
- For each observation it calculates the sum of the square of the difference between actual and predicted values and then it averages out these values.
- The squaring is necessary here to remove the negative signs.
- It also gives more weight to large differences.
- It is called Mean Square Error because you find the average error.
- MSE is a simple metric to evaluate regression family of algorithms.

- But it is the least useful metric.
- If the MSE value is higher the worst the model created is.
- But if it is very low, then the model is considered to be good model.
- If it is cose to 0, then the model is almost perfect, which is very rare.
- MSE is useful in scenarios where we have unexpected data values, that is very high or very low values.
- The disadvantage of this metric is, it can over estimate the model's inefficiency in case of noisy data or bad data.

Root Mean Square Error (RMSE):

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

- RMSE basically tells you how concentrated the data is around the line of best fit.
- By taking the square root of the mean square error, it brings the errors on the same scale as of the target values scale.
- That is RMSE will have the same scale of values as of the target variable.
- It penalizes large errors more.

Mean Absolute Error (MAE):

$$\mathrm{MAE} = \frac{1}{n}\sum_{j=1}^{n} |y_j - \hat{y}_j|$$

- Here we find the error as an average of the absolute of the difference between the actual values and the predicted values.
- It is a linear score which means that all the individual differences are weighted equally in the average.
- For example the difference between 30 and 0 will be thrice the difference between 10 and 0.
- This metric is not sensitive to outliers as Mean Square Error.
- It does not penalize the errors like MSE.

- MAE is used in functional areas like finance where 30$ error is 3 times greater than 10$ error.
- Whereas in MSE, 30$ error is considered 9 times greater than 10$ error.
- MAE is a bit easier than RMSE.
- If there is outliers in the data, then go for MAE, otherwise go for MSE.